

1. (7%) What is the solution to the **recurrence relation** $a_n = -3a_{n-1} + 10a_{n-2}$ if $a_0 = 3$ and $a_1 = 2$?

Ans:

此線性齊次遞迴關係之特徵方程式為 $r^2 + 3r - 10 = 0$

$\Rightarrow (r+5)(r-2) = 0 \Rightarrow r = -5$ 或 $r = 2$ (兩相異特徵根)

$\Rightarrow$ 其解的型式必為 $a_n = c(-5)^n + d(2)^n$, c, d 為常數

又 $a_0 = 3 = c + d$, $a_1 = 2 = -5c + 2d$, 解聯立方程式得 $c = \frac{4}{7}$, $d = \frac{17}{7}$

因此可得解為 $a_n = \frac{4}{7}(-5)^n + \frac{17}{7}2^n$

2. (7%) Solve $a_n = 4a_{n-1} - 4a_{n-2}$, $a_0 = 1$, $a_1 = 4$.

Ans:

此線性齊次遞迴關係之特徵方程式為 $r^2 - 4r + 4 = 0$

$\Rightarrow (r-2)^2 = 0 \Rightarrow r = 2$ (2重根)

$\Rightarrow$ 其解的型式必為 $a_n = c(2)^n + dn(2)^n$, c, d 為常數

又 $a_0 = 1 = c$, $a_1 = 4 = 2c + 2d$, 解聯立方程式得 $c = 1$, $d = 1$

因此可得解為 $a_n = (2)^n + n(2)^n$

3. (7%) Use the **inclusion-exclusion principle** to count the number of integers between 1 and 5,000 that are not divisible by 10, 12, or 21.

$$\begin{aligned} \text{Ans: } & 5000 - (\lfloor 5000/10 \rfloor + \lfloor 5000/12 \rfloor + \lfloor 5000/21 \rfloor) \\ & + (\lfloor 5000/\text{lcm}(10,12) \rfloor + \lfloor 5000/\text{lcm}(10,21) \rfloor + \lfloor 5000/\text{lcm}(12,21) \rfloor) \\ & - \lfloor 5000/\text{lcm}(10,12,21) \rfloor \\ & = 4000 \end{aligned}$$

4. (7%) Solve $a_n = 8a_{n-1} + 10^{n-1}$, $a_0 = 1$, using **generating functions** (限用生成函數).

$$\text{Ans: 令 } G(x) = \sum_{n=1}^{\infty} a_n x^n$$

$$\text{則依原遞迴關係} \Rightarrow \sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty} (8a_{n-1} + 10^{n-1}) x^n$$

$$\Rightarrow \sum_{n=0}^{\infty} a_n x^n - a_0 = 8x \sum_{n=1}^{\infty} a_{n-1} x^{n-1} + x \sum_{n=1}^{\infty} 10^{n-1} x^{n-1}$$

$$\Rightarrow G(x) - a_0 = 8x \sum_{n=0}^{\infty} a_n x^n + x \sum_{n=0}^{\infty} 10^n x^n$$

$$\Rightarrow G(x) - a_0 = 8xG(x) + x \sum_{n=0}^{\infty} 10^n x^n$$

$$\Rightarrow G(x) - 1 = 8xG(x) + x \frac{1}{1-10x}$$

$$\Rightarrow (1 - 8x)G(x) = x \frac{1}{1-10x} + 1 = \frac{x+1-10x}{1-10x} = \frac{1-9x}{1-10x}$$

$$\Rightarrow G(x) = \frac{1-9x}{(1-8x)(1-10x)} = \frac{1/2}{1-8x} + \frac{1/2}{1-10x} = \frac{1}{2} \left(\sum_{n=0}^{\infty} 8^n x^n + \sum_{n=0}^{\infty} 10^n x^n \right) = \sum_{n=0}^{\infty} \frac{1}{2} (8^n + 10^n) x^n$$

$$\text{故可得 } a_n = \frac{1}{2} (8^n + 10^n)$$

5. (7%) An alphabet set $\{A, B, C, D, E, F, G, H\}$ is used to form a string. A legal string **cannot** have AA, AB, BA, or BB in any part of the string. Suppose P_n is the number of legal strings of length $n+1$.

(a) (2%) Find P_0 and P_1 .

(a) (5%) Find the recurrence relation for P_n .

Ans: (a) $P_0=8$ and $P_1=60$ (b) $P_n = 6 P_{n-1} + 12 P_{n-2}$.

6. (7%) Use the **generating function** to determine the number of integer solutions to $a + b + c = 11$ where $2 \leq a \leq 4$, $3 \leq b \leq 5$ and $4 \leq c \leq 6$.

Ans: $a + b + c = 11$ 解的組數 $= (x^2 + x^3 + x^4)(x^3 + x^4 + x^5)(x^4 + x^5 + x^6)$ 展開後 x^{11} 項的係數共有以下6組解:

2	3	6
2	4	5
2	5	4
3	3	5
3	4	4
4	3	4

7. (5%) Determine the sequence whose **generating function** is $\frac{x}{2-x^2}$.

Ans: $0, \frac{1}{2}, 0, \frac{1}{4}, 0, \frac{1}{8}, 0, \frac{1}{16}, 0, \dots$

8. (5%) The hatchchecker at a restaurant neglects to place name tags on 6 hats. Each of the 6 customers is given a randomly selected hat upon exiting. Determine the number of possible cases in which **only one gets the correct hat**.

Ans: 重要觀念: 令 A_i 表集合 $\{1, 2, \dots, n\}$ 所有排列中恰有 i 個數字待在原位者. 則 $|A_i| = C(n, i) \times D_{n-i}$. 依題目 $n=6$, $|A_1| = C(6, 1) D_5 = 6 \times 44 = 264$

(Recall: $D_5 = 5! - C(5, 1)4! + C(5, 2)3! - C(5, 3)2! + C(5, 4)1! - C(5, 5)0! = 120 - 120 + 60 - 20 + 5 - 1 = 44$)

9. (5%) Let $X = \{A, B, C\}$ and $Y = \{1, 2, 3\}$. How many functions $f: X^2 \rightarrow Y$ are **onto**?

Ans: $3^9 - C(3, 1)2^9 + C(3, 2)1^9$

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

10. (15%) Given the matrix representation of the relation R , $M_R =$

(a) Is R **reflexive**?

(b) Is R **symmetric**?

(c) Is R **antisymmetric**?

(d) Is R **transitive**?

(e) Is R a **partial ordering** relation.

Ans: (a)Yes (b)No (c)Yes (d)no, 例(6,3),(3,1)應有(6,1)! (e)No

11. (15%) Consider the divisibility relation (i.e., $|$) and the poset $R = (S, |)$, where $S = \{1, 2, 3, 4, 6, 8, 12, 24\}$.
- Is R a **total order**?
 - Is R a **well order**?
 - Find all **maximal** elements.
 - Does R have a **greatest lower bound**? If yes, what's it?
 - Is R a **lattice**?

Ans: (a) No (b) No (c) 24 (d) yes, 1 (e) yes

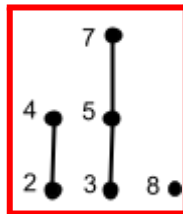
12. (20%) Let R_0 be the relation $\{(a, b) \mid b - a = 2\}$ on the set $A = \{2, 3, 4, 5, 7, 8\}$.
- Let R_1 is the reflexive closure of R_0 , Find R_1 .
 - Let R_2 is the symmetric closure of R_0 , Find R_2 .
 - Let R_3 is the transitive closure of R_0 , Find R_3 .
 - Among $R_1 \cup R_2$, $R_1 \cup R_3$ and $R_2 \cup R_3$, (i) determine which one is a **partial order** and (ii) draw its Hasse diagram.
 - Find **two** different **total orderings** that are compatible with the partial order found in (d).

Ans:

(a) $R_1 = \{(2,2), (3,3), (4,4), (5,5), (7,7), (8,8), (2,4), (3,5), (5,7)\}$

(b) $R_2 = \{(4,2), (5,3), (7,5), (2,4), (3,5), (5,7)\}$

(c) $R_3 = \{(3,7), (2,4), (3,5), (5,7)\}$



(d) (i) $R_1 \cup R_3$, (ii)

(e) 2,4,3,5,7,8 and 2,3,4,5,7,8 (答案不是唯二)

13. (7%) $\{\{1\}, \{2, 3, 4\}, \text{ and } \{5, 6\}\}$ is a partition of $A = \{1, 2, 3, 4, 5, 6\}$. Find the equivalence relation R such that R 's different equivalence classes form the partition.

Ans: $R = \{(1,1), (2,2), (3,3), (4,4), (2,3), (3,2), (2,4), (4,2), (3,4), (4,3), (5,5), (6,6), (5,6), (6,5)\}$